Exercise 2: SQL Aggregate functions & Operators

01. ~~SELECT distinct departments~~
~~Group by departments~~
~~FROM The students table;~~

SELECT distinct departments
FROM The students table
Group by departments;

department
IT
HR
FINANCE

02 SELECT ~~Av~~ avg age,
department
FROM The students table
GROUP by department;

department	av-age
IT	20,5
HR	22
FINANCE	23

03. SELECT departments,
student count
FROM The students table
Group by departments;

departments	student count
IT	2
HR	2

04 SELECT COUNT student_id,
name,
age,
department
FROM The students table
Group by age 21 and 23 >;

Student id	Name	age	department
02	Bob	22	HR
03	Charlie	21	IT
04	Diana	23	FINANCE
05	Eve	22	HR

05. SELECT count id as no students
IT,
HR,
FROM The students table
Group by age 21 >;

Student id	Name	age	department
02	Bob	22	HR
03	Charlie	21	IT
05	Eve	22	HR

06. SELECT count credits,
department
FROM The courses table
Group by department 5 >;

department	total credits
IT	8

07 SELECT ~~...~~ count id,
course_name,
department,
credits
~~...~~ FROM The courses table
Group by credits <3;

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

08 SELECT count_id,
course_name,
credits
FROM The courses table
Group by credits descending order;

courses_id	course_name	credits
~~...~~ 102	Python	4
~~103~~ 103	Data science	4
105	Statistics	3

09. SELECT maximum,
minimum,
average grade (across all enrollments)
From The enrollments table
Group by enrollments;

max_grade	min_grade	avg_grade
90	78	84,6

10. SELECT count enrollment_id, (sum above "count")
course_id
From The enrollment table
Group by ~~courses~~ per course;

course_id	enrollment_count
1	101
2	102
3	103
4	104
5	105

11 SELECT ~~sa~~ sum salary,
bonus
FROM The salaries table
Group by per department;

department	total salary	total bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	62 000	6 000

12 SELECT sum departments,
AVG_salary
From The salaries table
Group by 55 000>;

department	avg_salary
IT	61 000
FINANCE	70 000

13 SELECT sum employee_id,
name,
salary,
bonus, total compensation.
From The salaries table
Group by 60,000>;

empolyee_id	name	department	salary	bonus	total comp
1	Tom	~~IT~~	60 000	5 000	65000
3	Spike	~~Finance~~	70 000	6000	76000
4	Tyke	~~IT~~	62 000	5500	67500

14. SELECT sum of budget,
avg_budget,
department
From The projects table
Grop by ~~departments~~; ~~budget~~ average budget 70 000>;

department	total budget	avg_budget
~~IT~~	~~27000~~	
~~Finance~~	~~80 000~~	
~~Marketing~~	~~60 000~~	
~~HR~~	~~50 000~~	

department	total budget	avg_budget
IT	27 0000	135 000
Finance	80 000	80000

15. SELECT projects_id,
project-name,
department,
budget
FROM The projects table
Group by budgets between 50,000 and 120,000;

project_id	project name	department	budget
1	AI APP	IT	120000
2	Payroll System	Finance	80 000
4	Website	Marketing	60 000
5	HR Portal	HR	50 000